

Brian Ojoh Mile

📍 Laurel, Maryland 📞 (240) 524-0631 @ brianmile10@yahoo.com

Summary

Full Stack Engineer building scalable intelligence platforms for government and enterprise. Delivered geospatial systems unifying 35+ real-time data layers and architected distributed ETL pipelines spanning 40+ machines. Strong foundation in high-performance vector tile backends, real-time WebSocket systems, and large-scale data processing.

Experience

ICYLON LLC

May 2024 - December 2025

Full Stack Software Engineer

Vienna, Virginia

- Architected and built the framework for a collaborative Agentic AI platform in React/Next.js, featuring an AI Orchestrator that generates complete data workflows from natural language. Collaborated with others to implement 60+ specialized nodes spanning multi-domain intelligence, social media, telecom lookups, and AI-powered research (RAG, report generation, facial recognition). Platform was recognized as awardable for contract by Tradewinds Solutions Marketplace.
- Developed and launched a drag-and-drop ETL workflow platform in React/Next.js using a node-based canvas (React Flow), enabling analysts to build pipelines across CSV, PostgreSQL, and MongoDB with 12+ configurable transformations (column ops, regex replacement, date normalization), automated graph layout, AI-assisted processing nodes, and real-time job state synchronization via WebSockets.
- Engineered a distributed ETL execution engine using Flask, Celery, and a Dask cluster spanning 40+ machines, enabling large-scale CSV and PostgreSQL processing with configurable transformations, automated file-driven pipeline triggering, Parquet-based storage with intelligent partitioning, and real-time job state propagation via WebSockets.
- Designed and delivered a production geospatial intelligence platform using React/Next.js and MapLibre, unifying 35+ real-time data layers—including military installations, maritime vessels, flight tracking, power plants, and transportation networks—into a single interactive map featuring real-time tracking, temporal analysis with configurable time windows, advanced search, persistent filters, and geospatial measurement tools; implemented a modular frontend architecture that reduced onboarding of new intelligence layers from weeks to days and significantly improved analyst workflows.
- Built a high-performance vector tile backend using PostGIS with automated SQL generation, materialized views (spatial indexing, concurrent refresh), multi-source geospatial ingestion (OSM, GeoJSON, MongoDB), and Redis caching to support scalable, low-latency map serving across dozens of layers. Enabled declarative, YAML-based configuration of tile schemas and zoom-dependent behavior to reduce iteration time and operational complexity.
- Designed and deployed a full-stack support request management system, streamlining customer interactions and improving satisfaction.
- Secured backend infrastructure by migrating Neo4j connections from direct UI access to a Flask-based API, eliminating credential exposure on the frontend. This transition enforced read-only queries, improved concurrency handling, and provided a structured API for a React-based frontend.

DATA ANNOTATION TECH

October 2021 - May 2024

AI Trainer

Remote

- Created programming challenges and walkthroughs for LLM training, focusing on data analysis, visualization, and algorithms.
- Evaluated AI-generated code for performance, correctness, and security; provided feedback loops to improve model output quality.
- Reviewed and rated contributions from other trainers to uphold dataset consistency and engineering standards.

Education

University of Maryland

2017-2021

Bachelor's in Computer Science

College Park, Maryland

Skills

Languages: Python, TypeScript/JavaScript, HTML/CSS, Kotlin, Java, C, Ruby, Rust

Frontend: React, Svelte, NextJS, Redux, Zustand, TailwindCSS, Material UI, Deck.gl, Maplibre.gl, ReactFlow

Backend & APIs: Flask, FastAPI, NodeJS, Express

Data & Infrastructure: MongoDB, PostgreSQL, Pandas, Dask, Neo4j, GridFS, REST/GraphQL APIs

Developer Certifications (FreeCodeCamp): Foundational C#, Backend Development and API's, Javascript Algorithms and Data Structures, Responsive Web Design